

DEPARTMENT OF MATHEMATICS, IIT - GUWAHATI

Odd Semester of the Academic Year 2022-2023

MA 101 Mathematics II

Problem Sheet 2: Partial derivatives, tangent and normals, differentials, gradient, directional derivatives and chain rules etc.

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1. Let $f(x, y) = \begin{cases} \frac{x^2 - xy}{x+y} & \text{if } x + y \neq 0 \\ 0 & \text{if } x + y = 0. \end{cases}$

Find

(a) $f_x(0, 0), f_y(0, 0)$

(b) $\lim_{(x,y) \rightarrow (0,0)} f_x(x, y)$, if it exists and check whether it is equal to $f_x(0, 0)$.

Solution: Note that $f_x(0, 0)$ if it exists, is given by:

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h - 0}{h} = 1.$$

Similarly $f_y(0, 0)$ if it exists, is given by:

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{0}{k} = 0.$$

(b) Also $f_x(x, y)$ (for $x + y \neq 0$) if it exists, is given by:

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}.$$

Since (x, y) is such that $x + y \neq 0$, if h sufficiently small such that $x + y + h \neq 0$, then

$$\begin{aligned} \frac{f(x+h, y) - f(x, y)}{h} &= \frac{\frac{(x+h)^2 - (x+h)y}{(x+h)+y} - \frac{x^2 - xy}{x+y}}{h} = \frac{hx^2 + 2hxy - hy^2 + h^2x + h^2y}{h(x+y)(x+h+y)} \\ &= \frac{x^2 + 2xy - y^2 + hx + hy}{(x+y)(x+h+y)}. \end{aligned}$$

Since $\lim_{h \rightarrow 0} x^2 + 2xy - y^2 + hx + hy = x^2 + 2xy - y^2$ and

$\lim_{h \rightarrow 0} (x+y)(x+h+y) = (x+y)^2 \neq 0$ for $x + y \neq 0$,

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} = \frac{x^2 + 2xy - y^2}{(x+y)^2} = 1 - \frac{2y^2}{(x+y)^2},$$

for $x + y \neq 0$.

Clearly $\lim_{(x,y) \rightarrow (0,0)} f_x(x, y) = \lim_{(x,y) \rightarrow (0,0)} 1 - \frac{2y^2}{(x+y)^2},$

does not exist (consider $(x, y) \rightarrow (0, 0)$ such that $y = mx, m \neq -1$).

2. Let $f(x, y) = \sqrt{x^2 + y^2}$.

(a) Find $f_x(x, y)$ and $f_y(x, y)$ for $(x, y) \neq (0, 0)$

(b) Show that $f_x(0, 0)$ and $f_y(0, 0)$ do not exist.

Solution: (a) Note that $f_x(x, y)$ if it exists, is given by

$$\begin{aligned} f_x(x, y) &= \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \\ &= \frac{f(x+h, y) - f(x, y)}{h} = \frac{\sqrt{(x+h)^2 + y^2} - \sqrt{x^2 + y^2}}{h} \\ &= \frac{(\sqrt{(x+h)^2 + y^2} - \sqrt{x^2 + y^2})(\sqrt{(x+h)^2 + y^2} + \sqrt{x^2 + y^2})}{h(\sqrt{(x+h)^2 + y^2} + \sqrt{x^2 + y^2})} = \frac{h + 2x}{\sqrt{(x+h)^2 + y^2} + \sqrt{x^2 + y^2}}, \end{aligned}$$

for $(x, y) \neq (0, 0)$.

$$\text{Hence } f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} = \frac{x}{\sqrt{x^2 + y^2}}, \text{ for } (x, y) \neq (0, 0).$$

$$\text{Similarly } f_y(x, y) = \lim_{k \rightarrow 0} \frac{f(x, y+k) - f(x, y)}{k} = \frac{y}{\sqrt{x^2 + y^2}}, \text{ for } (x, y) \neq (0, 0).$$

Aliter: $f_x(x, y)$ for $(x, y) \neq (0, 0)$ may be obtained directly (without going through first principles) by taking the single variable derivative of $f(x, y)$ with respect to x by treating y as constant.

Similarly $f_y(x, y)$ for $(x, y) \neq (0, 0)$ may be obtained directly (without going through first principles) by taking the single variable derivative of $f(x, y)$ with respect to y by treating x as constant.

$$(b) \text{ Also } f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h}, \text{ does not exist.}$$

Similarly $f_y(0, 0)$ does not exist.

3. Let

$$f(x, y) = \begin{cases} \frac{xy^3}{x^2 + y^6} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

(a) Calculate $f_x(x, y)$ and $f_y(x, y)$ at all points where $(x, y) \neq (0, 0)$.

(b) Compute all first and second order partial derivatives at $(0, 0)$ if they exist.

(c) Show that f is discontinuous at $(0, 0)$.

Solution: (a)

$$f_x(x, y) = \frac{y^3(y^6 - x^2)}{(x^2 + y^6)^2} \text{ and } f_y(x, y) = \frac{3xy^2(x^2 - y^6)}{(x^2 + y^6)^2}, \text{ for } (x, y) \neq (0, 0). \quad (1)$$

$$f_x(x, 0) = \lim_{h \rightarrow 0} \frac{f(x+h, 0) - f(x, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0 \text{ for } x \neq 0.$$

$$f_y(0, y) = \lim_{k \rightarrow 0} \frac{f(0, y+k) - f(0, y)}{k} = \lim_{k \rightarrow 0} \frac{0}{k} = 0 \text{ for } y \neq 0.$$

$$f_y(x, 0) = \lim_{k \rightarrow 0} \frac{f(x, k) - f(x, 0)}{k} = \lim_{k \rightarrow 0} \frac{\frac{xk^3}{x^2 + k^6} - 0}{k} = \lim_{k \rightarrow 0} \frac{xk^2}{x^2 + k^6} = 0 \text{ for } x \neq 0.$$

$$f_x(0, y) = \lim_{h \rightarrow 0} \frac{f(h, y) - f(0, y)}{h} = \lim_{h \rightarrow 0} \frac{\frac{hy^3}{h^2 + y^6} - 0}{h} = \frac{y^3}{y^6} = \frac{1}{y^3} \text{ for } y \neq 0.$$

Aliter: The above expressions for

$$f_x(x, 0) = 0 \text{ for } x \neq 0,$$

$$\begin{aligned}
f_y(0, y) &= 0 \text{ for } y \neq 0, \\
f_y(x, 0) &= 0 \text{ for } x \neq 0, \\
f_x(0, y) &= \frac{1}{y^3} \text{ for } y \neq 0,
\end{aligned}$$

could be obtained directly by suitable substitutions in (1).

$$(b) \ f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0.$$

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{0}{k} = 0.$$

$$\text{Hence } f_{xx}(0, 0) = \lim_{h \rightarrow 0} \frac{f_x(h, 0) - f_x(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0.$$

$$f_{yy}(0, 0) = \lim_{k \rightarrow 0} \frac{f_y(0, k) - f_y(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{0}{k} = 0.$$

$$f_{xy}(0, 0) = \lim_{k \rightarrow 0} \frac{f_x(0, k) - f_x(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{\frac{1}{k^3} - 0}{k} = \lim_{k \rightarrow 0} \frac{1}{k^4}, \text{ does not exist.}$$

$$f_{yx}(0, 0) = \lim_{h \rightarrow 0} \frac{f_y(h, 0) - f_y(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0.$$

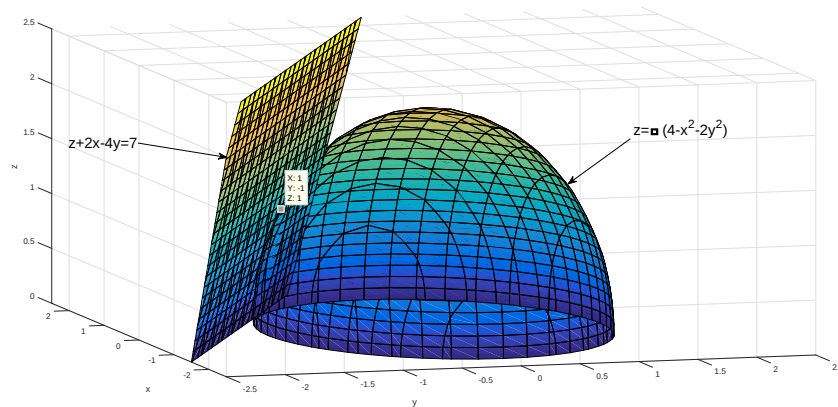
(c) We had already seen (tut-1, 14(d)) that $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist, hence f is not continuous at $(0, 0)$.

4. Find the equation of the tangent plane to the surface $z = \sqrt{4 - x^2 - 2y^2}$ at the point $(1, -1, 1)$.

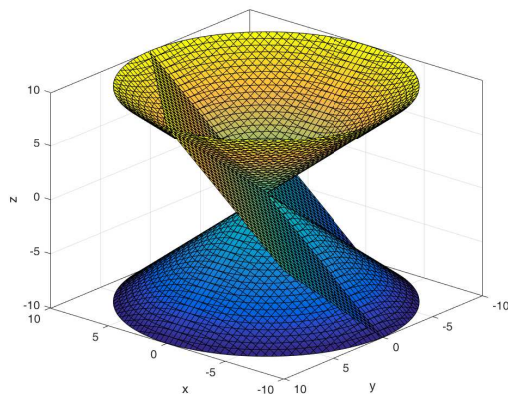
Solution: $(z - z_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$

$$\Rightarrow (z - z_0) = \left(\frac{-x_0}{\sqrt{4 - x_0^2 - 2y_0^2}} \right) (x - x_0) + \left(\frac{-2y_0}{\sqrt{4 - x_0^2 - 2y_0^2}} \right) (y - y_0)$$

$$\Rightarrow (z - 1) = (-1)(x - 1) + (2)(y + 1).$$



5. It is geometrically evident that every plane tangent to the cone $z^2 = x^2 + y^2$ pass through the origin. Show this by the method of calculus.



Solution: Since the level surface is of the form $f(x, y, z) = 0$, where $f(x, y, z) = z^2 - (x^2 + y^2)$, the tangent plane of the level surface at (x_0, y_0, z_0) is of the form:
 $(z - z_0)f_z(x_0, y_0, z_0) + (x - x_0)f_x(x_0, y_0, z_0) + (y - y_0)f_y(x_0, y_0, z_0) = 0$.
 $\Rightarrow (z - z_0)(2z_0) + (x - x_0)(-2x_0) + (y - y_0)(-2y_0) = 0$,
 $\Rightarrow zz_0 - xx_0 - yy_0 = 0$ since $z_0^2 - x_0^2 - y_0^2 = 0$.
Clearly $(x, y, z) = (0, 0, 0)$ satisfies the above equation for all points (x_0, y_0, z_0) of the cone.

6. Find the equations of the tangent plane and normal line to the given surface at the specified point

(a) $x^2 + y^2 - z^2 - 2xy + 4xz = 4$, $(1, 0, 1)$.

(b) $z + 1 = xe^y \cos z$, $(1, 0, 0)$.

Solution:

(a) Since the level surface is of the form $f(x, y, z) = 4$, where $f(x, y, z) = x^2 + y^2 - z^2 - 2xy + 4xz$, the tangent plane of the level surface at $(x_0, y_0, z_0) = (1, 0, 1)$ is of the form:
 $(z - z_0)f_z(x_0, y_0, z_0) + (x - x_0)f_x(x_0, y_0, z_0) + (y - y_0)f_y(x_0, y_0, z_0) = 0$. (1)
where $f_x(x_0, y_0, z_0) = 2x_0 - 2y_0 + 4z_0$, $f_y(x_0, y_0, z_0) = 2y_0 - 2x_0$ and $f_z(x_0, y_0, z_0) = -2z_0 + 4x_0$.

(1) implies $\Rightarrow (z - 1)2 + (x - 1)6 + y(-2) = 0$ or $2z + 6x - 2y = 8$.

The symmetric equations of the normal line is given by:

$$\frac{(z - z_0)}{f_z(x_0, y_0, z_0)} = \frac{(x - x_0)}{f_x(x_0, y_0, z_0)} = \frac{(y - y_0)}{f_y(x_0, y_0, z_0)}, \quad \text{or}$$

$$\frac{(z - 1)}{2} = \frac{(x - 1)}{6} = \frac{(y - 0)}{-2}.$$

(b) Since the level surface is of the form $f(x, y, z) = -1$,

where $f(x, y, z) = z - xe^y \cos z$,

the tangent plane of the level surface at $(x_0, y_0, z_0) = (1, 0, 0)$ is of the form:

$(z - z_0)f_z(x_0, y_0, z_0) + (x - x_0)f_x(x_0, y_0, z_0) + (y - y_0)f_y(x_0, y_0, z_0) = 0$, (1)

where $f_x(x_0, y_0, z_0) = -e^{y_0} \cos z_0$, $f_y(x_0, y_0, z_0) = -x_0 e^{y_0} \cos z_0$

and $f_z(x_0, y_0, z_0) = 1 + x_0 e^{y_0} \sin z_0$.

$$(1) \Rightarrow (z - 0)1 + (x - 1)(-1) + (y - 0)(-1) = 0 \text{ or } z - x - y = -1.$$

The symmetric equations of the normal line is given by:

$$\frac{(z - z_0)}{f_z(x_0, y_0, z_0)} = \frac{(x - x_0)}{f_x(x_0, y_0, z_0)} = \frac{(y - y_0)}{f_y(x_0, y_0, z_0)}$$

$$\frac{(z - 0)}{1} = \frac{(x - 1)}{-1} = \frac{(y - 0)}{-1}.$$

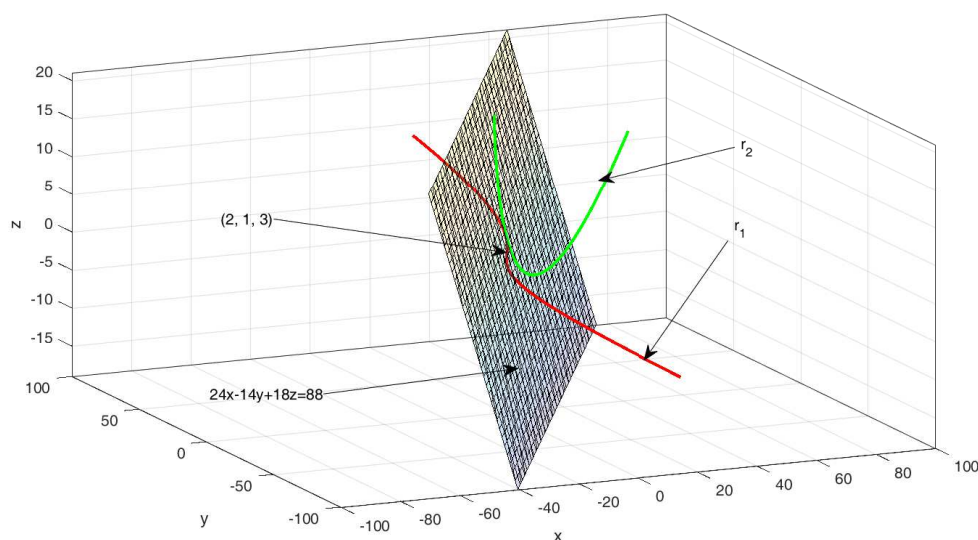
7. Suppose we are seeking the equation of the tangent plane to a surface S at the point $P = (2, 1, 3)$. You don't have an equation for S , but you know that the curves

$$\mathbf{r}_1(t) = \langle 2 + 3t, 1 - t^2, 3 - 4t + t^2 \rangle$$

$$\mathbf{r}_2(t) = \langle 1 + t^2, 2t^3 - 1, 2t + 1 \rangle$$

both lie on S . Find an equation of the tangent plane at P .

Solution: Clearly $\mathbf{r}_1(0)$ corresponds to $(2, 1, 3)$, $\mathbf{r}_2(1)$ corresponds to $(2, 1, 3)$.



Also the tangent plane to the surface S at $(2, 1, 3)$ contains the tangent lines to each of the curves $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ at $(2, 1, 3)$.

The tangent vectors at $(2, 1, 3)$ for $\mathbf{r}_1, \mathbf{r}_2$ at $(2, 1, 3)$ are given by:

$$\mathbf{r}'_1(0) = \langle 3, 0, -4 \rangle,$$

$$\mathbf{r}'_2(1) = \langle 2, 6, 2 \rangle,$$

respectively.

Hence a normal to the surface is one which is orthogonal to both these tangent vectors and is given by:

$$\mathbf{r}'_1(0) \times \mathbf{r}'_2(1) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 0 & -4 \\ 2 & 6 & 2 \end{vmatrix} = \langle 24, -14, 18 \rangle.$$

Hence an equation of the tangent plane is given by:

$$24(x - 2) + (-14)(y - 1) + 18(z - 3) = 0.$$

8. Show that the sum of the x -, y -, and z -intercepts of any tangent plane (at any point of the surface wherever it is defined) to the surface $\sqrt{x} + \sqrt{y} + \sqrt{z} = \sqrt{c}$ is a constant.

Solution: The equation of the tangent plane at (x_0, y_0, z_0) is given by

$$(z - z_0)f_z(x_0, y_0, z_0) + (x - x_0)f_x(x_0, y_0, z_0) + (y - y_0)f_y(x_0, y_0, z_0) = 0.$$

$$\Rightarrow \frac{(z - z_0)}{2\sqrt{z_0}} + \frac{(x - x_0)}{2\sqrt{x_0}} + \frac{(y - y_0)}{2\sqrt{y_0}} = 0,$$

provided none of x_0, y_0, z_0 is equal to 0.

$$\Rightarrow \frac{z}{\sqrt{z_0}} + \frac{x}{\sqrt{x_0}} + \frac{y}{\sqrt{y_0}} = (\sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0}) = \sqrt{c}.$$

$$\Rightarrow \frac{x}{\sqrt{cx_0}} + \frac{y}{\sqrt{cy_0}} + \frac{z}{\sqrt{cz_0}} = 1.$$

Hence the sum of the intercepts is

$$(\sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0})\sqrt{c} = c.$$

9. If $z = f(x, y) = x^2 + 3xy - y^2$,

- write the expression for the differential dz at (x, y, z) ;
- and if x changes from 2 to 2.05 and y changes from 3 to 2.96, compare the values of Δz and dz .

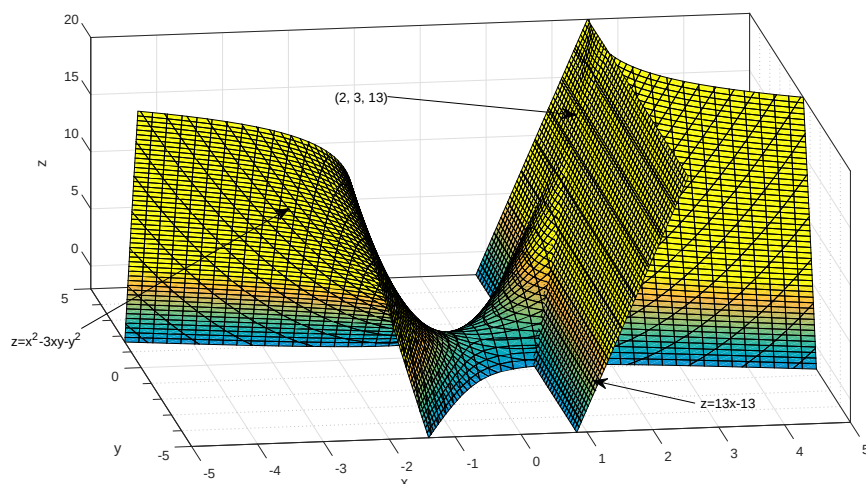
Solution:

- (a) The differential dz at (x, y, z) is given by:

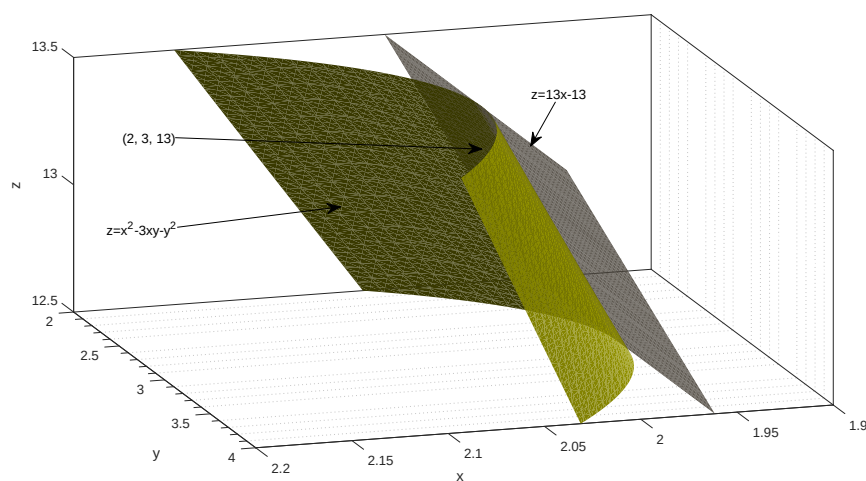
$$\begin{aligned} dz &= \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy \\ &= (2x + 3y)dx + (3y - 2x)dy. \end{aligned}$$

- (b) Now $dx = \Delta x = 2.05 - 2 = 0.05$, $dy = \Delta y = -3 + 2.96 = -0.04$.

$$\begin{aligned} \text{Hence at } (2, 3, f(2, 3)), \quad dz &= (2 \times 2 + 3 \times 3)(0.05) + (3 \times 2 - 2 \times 3)(-0.04) = 0.65 \\ \Delta z &= f(2.05, 2.96) - f(2, 3) = 0.6449. \end{aligned}$$



(a)



(b)

10. Find the directional derivative of the function at the given point in the direction of the vector $\mathbf{v}$.

(a) $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$, $(1, 2, -2)$, $\mathbf{v} = \langle -6, 6, -3 \rangle$

(b) $g(x, y, z) = x \tan^{-1}\left(\frac{y}{z}\right)$, $(1, 2, -2)$, $\mathbf{v} = \mathbf{i} + \mathbf{j} - \mathbf{k}$.

Solution: (a) The directional derivative of f at $(1, 2, -2)$ along the unit vector $\mathbf{u} = \left\langle -\frac{2}{3}, \frac{2}{3}, -\frac{1}{3} \right\rangle$ if it exists, is equal to the following limit:

$$\begin{aligned} D_{\mathbf{u}}f(1, 2, -2) &= \lim_{t \rightarrow 0} \frac{f\left((1, 2, -2) + t\left(-\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right)\right) - f(1, 2, -2)}{t} \\ &= \frac{f\left((1, 2, -2) + t\left(-\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right)\right) - f(1, 2, -2)}{t} = \frac{\sqrt{\left(1 - \frac{2}{3}t\right)^2 + \left(2 + \frac{2}{3}t\right)^2 + \left(-2 - \frac{1}{3}t\right)^2} - 3}{t} \\ &= \frac{\sqrt{9 + t^2 + \frac{8}{3}t} - 3}{t}. \end{aligned}$$

By applying L'Hospital's rule we get the limit as

$$= \lim_{t \rightarrow 0} \frac{2t + \frac{8}{3}}{2\sqrt{9 + t^2 + \frac{8}{3}t}} = \frac{4}{9}.$$

(b) The directional derivative of f at $(1, 2, -2)$ along the unit vector $\mathbf{u} = \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right\rangle$ is equal to the following limit, if it exists.

$$\begin{aligned} D_{\mathbf{u}}f(1, 2, -2) &= \lim_{t \rightarrow 0} \frac{f((1, 2, -2) + t(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})) - f(1, 2, -2)}{t} \\ &= \frac{f((1, 2, -2) + t(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})) - f(1, 2, -2)}{t} \\ &= \frac{(1 + \frac{t}{\sqrt{3}})\tan^{-1}(-1) - \tan^{-1}(-1)}{t}, \text{ for } t \neq 0 \text{ and } t \text{ sufficiently small.} \\ &= \frac{\frac{t}{\sqrt{3}}\tan^{-1}(-1)}{t} = \frac{1}{\sqrt{3}}\tan^{-1}(-1) = -\frac{\pi}{4\sqrt{3}}, \\ &\Rightarrow \lim_{t \rightarrow 0} \frac{f((1, 2, -2) + t(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})) - f(1, 2, -2)}{t} = -\frac{\pi}{4\sqrt{3}}. \end{aligned}$$

Aliter: Note that if the existence of all directional derivatives of f at (x_0, y_0, z_0) is already guaranteed (say when f is differentiable at (x_0, y_0, z_0)) then one can directly calculate the value of the directional derivative by the formula given below:

$$D_{\mathbf{u}}f(x_0, y_0, z_0) = [\nabla f]_{(x_0, y_0, z_0)} \cdot \mathbf{u}.$$

11. Find the directional derivatives of the scalar field $f(x, y) = x^3 - 3xy$ along the parabola $y = x^2 - x + 2$ at the point $(1, 2)$.

Solution: A unit vector along the tangent to the parabola at $(1, 2)$ can be obtained as follows:

The slope of the tangent vector of the parabola at $(1, 2)$ is given by

$$\frac{dy}{dx} \Big|_{x=1} = (2x - 1)_{x=1} = 1.$$

Hence the tangent line of the parabola at $(1, 2)$ is given by $y - 2 = 1(x - 1)$ or $y - x = 1$.

Any two points on this tangent line can be taken as $P_0(x_0, 1+x_0)$ and $P_1(2x_0, 1+2x_0)$ (you can choose any arbitrary points in such a way that $y = 1 + x$).

Therefore the vector $\overrightarrow{P_0P_1}$ joining the points P_0 and P_1 is given by

$\langle 2x_0 - x_0, (1+2x_0) - (1+x_0) \rangle = \langle x_0, x_0 \rangle$ so that a unit vector along this tangent

is $\left\langle \frac{x_0}{\sqrt{x_0^2 + x_0^2}}, \frac{x_0}{\sqrt{x_0^2 + x_0^2}} \right\rangle = \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$. (You can also obtain the unit tangent by

parametrizing the parabola by the vector $\mathbf{r}(t) = \langle t, t^2 - t + 1 \rangle$, so that a unit vector at the point $(1, 2)$ which corresponds to $t = 1$ is given by $\hat{u} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} \Big|_{t=1} =$

$$\left\langle \frac{1}{\sqrt{1 + (2t - 1)^2}}, \frac{2t - 1}{\sqrt{1 + (2t - 1)^2}} \right\rangle \Big|_{t=1} = \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle.$$

Note that depending upon the orientation of the parabola, you may take this unit

vector as $\left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$ or $\left\langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right\rangle$.)

Since f is clearly differentiable throughout $\mathbf{R}^2$ (* the partial derivatives $f_x = 3x^2 - 3y$ and $f_y = -3x$ are continuous throughout $\mathbf{R}^2$), therefore the directional derivatives exist along any direction $\mathbf{u}$ and is given by

$$D_{\mathbf{u}}f(1,2) = [\nabla f]_{(1,2)} \cdot \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle = \langle -3, -3 \rangle \cdot \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle = -3\sqrt{2}.$$

(* For any $(x_0, y_0) \in \mathbf{R}^2$

$$\begin{aligned} |f_x(x, y) - f_x(x_0, y_0)| &= 3|(x^2 - x_0^2) - (y - y_0)| \leq 3(|x^2 - x_0^2| + |y - y_0|) \\ &= 3(|x - x_0||x + x_0| + |y - y_0|). \end{aligned}$$

Since (x, y) is sufficiently close to (x_0, y_0) , we can assume that $|x - x_0| < 1$, then $|x + x_0| < 2|x_0| + 1 = M$, say.

$$\text{Then } |f_x(x, y) - f_x(x_0, y_0)| < 3(M|x - x_0| + |y - y_0|) \leq (3M + 1)\sqrt{(x - x_0)^2 + (y - y_0)^2}$$

$$\text{Hence given } \epsilon > 0 \text{ take } \delta = \frac{\epsilon}{3M + 1},$$

then $|f_x(x, y) - f_x(x_0, y_0)| < \epsilon$ if $\sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$, which implies f_x is continuous at (x_0, y_0) .

Similarly one can show that f_y is continuous at (x_0, y_0) .)

12. Show that if the function $z = f(x, y)$ is differentiable at $\mathbf{x}_0 = \langle x_0, y_0 \rangle$, then

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} \frac{f(\mathbf{x}) - f(\mathbf{x}_0) - \nabla f(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0)}{|\mathbf{x} - \mathbf{x}_0|} = 0.$$

Solution: Since $z = f(x, y)$ is differentiable at $\mathbf{x}_0 = \langle x_0, y_0 \rangle$, by the definition of differentiability, we have

$$\Delta z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \epsilon_1\Delta x + \epsilon_2\Delta y \quad \dots (A),$$

where $\epsilon_1, \epsilon_2 \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$. Now, $\Delta z = f(\mathbf{x}) - f(\mathbf{x}_0)$, $\langle \Delta x, \Delta y \rangle = \mathbf{x} - \mathbf{x}_0$, so $\langle \Delta x, \Delta y \rangle \rightarrow (0, 0)$ is equivalent to $\mathbf{x} \rightarrow \mathbf{x}_0$ and $\langle f_x(x_0, y_0), f_y(x_0, y_0) \rangle = \nabla f(\mathbf{x}_0)$. Substituting in (A), we get

$$f(\mathbf{x}) - f(\mathbf{x}_0) = \nabla f(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0) + \langle \epsilon_1, \epsilon_2 \rangle \cdot \langle \Delta x, \Delta y \rangle, \quad \text{or}$$

$$\langle \epsilon_1, \epsilon_2 \rangle \cdot (\mathbf{x} - \mathbf{x}_0) = f(\mathbf{x}) - f(\mathbf{x}_0) - \nabla f(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0), \quad \text{and so}$$

$$\frac{f(\mathbf{x}) - f(\mathbf{x}_0) - \nabla f(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0)}{|\mathbf{x} - \mathbf{x}_0|} = \frac{\langle \epsilon_1, \epsilon_2 \rangle \cdot (\mathbf{x} - \mathbf{x}_0)}{|\mathbf{x} - \mathbf{x}_0|}. \quad \text{But } \frac{\mathbf{x} - \mathbf{x}_0}{|\mathbf{x} - \mathbf{x}_0|} \text{ is a unit}$$

vector, therefore $\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} \frac{\langle \epsilon_1, \epsilon_2 \rangle \cdot (\mathbf{x} - \mathbf{x}_0)}{|\mathbf{x} - \mathbf{x}_0|} = 0$, since $\epsilon_1, \epsilon_2 \rightarrow 0$ as $\mathbf{x} \rightarrow \mathbf{x}_0$. Hence

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} \frac{f(\mathbf{x}) - f(\mathbf{x}_0) - \nabla f(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0)}{|\mathbf{x} - \mathbf{x}_0|} = 0$$

13. Let $f(x, y) = \frac{x}{|x|} \sqrt{x^2 + y^2}$ if $x \neq 0$ and $f(x, y) = 0$ if $x = 0$. Show that f is continuous at $(0, 0)$ and the directional derivatives exist thereat, but it is not differentiable at $(0, 0)$.

Solution: $|f(x, y) - f(0, 0)| = |f(x, y)| = \frac{|x| \sqrt{x^2 + y^2}}{|x|} = \sqrt{x^2 + y^2}$ if $x \neq 0$
 $= 0$ if $x = 0$.

Given any $\epsilon > 0$ take $\delta = \epsilon$, then

$$|f(x, y) - f(0, 0)| \leq \sqrt{x^2 + y^2} < \epsilon \text{ if } \sqrt{x^2 + y^2} < \delta.$$

Hence f is continuous at $(0, 0)$.

The directional derivative of f at $(0, 0)$ along the unit vector $\mathbf{u} = \langle u_1, u_2 \rangle$, if it exists, is given by:

$$\begin{aligned} D_{\mathbf{u}}f(0, 0) &= \lim_{t \rightarrow 0} \frac{f(tu_1, tu_2) - f(0, 0)}{t} \\ \frac{f(tu_1, tu_2) - f(0, 0)}{t} &= \frac{tu_1 \sqrt{t^2 u_1^2 + t^2 u_2^2}}{|tu_1| t} = \frac{u_1}{|u_1|} \text{ if } u_1 \neq 0. \\ \frac{f(tu_1, tu_2) - f(0, 0)}{t} &= 0 \text{ if } u_1 = 0. \end{aligned}$$

Hence the directional derivatives exist along every $\mathbf{u} = \langle u_1, u_2 \rangle$ at $(0, 0)$.

To check the differentiability of f at $(0, 0)$ we need to check that

$$\begin{aligned} \lim_{(h_1, h_2) \rightarrow (0, 0)} \frac{f(h_1, h_2) - f_x(0, 0)h_1 - f_y(0, 0)h_2}{\sqrt{h_1^2 + h_2^2}} &= 0 \\ \frac{f(h_1, h_2) - f_x(0, 0)h_1 - f_y(0, 0)h_2}{\sqrt{h_1^2 + h_2^2}} &= \frac{\frac{h_1 \sqrt{h_1^2 + h_2^2}}{|h_1|} - h_1}{\sqrt{h_1^2 + h_2^2}}, \text{ if } h_1 \neq 0. \end{aligned}$$

If $h_1 = h_2 \neq 0$

$$\text{then } \frac{f(h_1, h_2) - f_x(0, 0)h_1 - f_y(0, 0)h_2}{\sqrt{h_1^2 + h_2^2}} = \left(1 - \frac{1}{\sqrt{2}}\right) \left(\frac{h_1}{|h_1|}\right).$$

$$\text{Hence } \lim_{(h_1, h_2) \rightarrow (0, 0)} \frac{f(h_1, h_2) - f_x(0, 0)h_1 - f_y(0, 0)h_2}{\sqrt{h_1^2 + h_2^2}} \text{ does not exist.}$$

Hence f is not differentiable at $(0, 0)$.

14. Show that the following functions are differentiable at the respective points mentioned below:

(a) Let

$$f(x, y) = \begin{cases} \frac{x}{x+y} & \text{if } x+y \neq 0 \\ 0 & \text{if } x+y = 0. \end{cases}$$

Show that f is differentiable at $(2, 1)$ but not differentiable at $(0, 0)$.

(b) Show that $f(x, y) = \sqrt{x + e^y}$ is differentiable at $(3, 0)$, where x, y is such that $x + e^y \geq 0$.

Solution:

(a) f is differentiable at $(2, 1)$ if

$$\lim_{(h, k) \rightarrow \mathbf{0}} \frac{|f(2+h, 1+k) - f(2, 1) - [\nabla f]_{(2,1)} \cdot \langle h, k \rangle|}{\sqrt{h^2 + k^2}} = 0.$$

$$\begin{aligned} & \left| \frac{f(2+h, 1+k) - f(2, 1) - [\nabla f]_{(2,1)} \cdot \langle h, k \rangle}{\sqrt{h^2 + k^2}} \right| = \left| \frac{\frac{2+h}{3+h+k} - \frac{2}{3} - \frac{h}{9} + \frac{2k}{9}}{\sqrt{h^2 + k^2}} \right| \\ &= \frac{|hk + 2k^2 - h^2|}{9(3+h+k)\sqrt{h^2 + k^2}} \leq \frac{|hk + 2k^2 - h^2|}{\sqrt{h^2 + k^2}} \text{ for sufficiently small } h \text{ and } k. \end{aligned}$$

$$\text{Hence the above expression is } \leq \frac{|hk|}{\sqrt{h^2 + k^2}} + \frac{2k^2}{\sqrt{h^2 + k^2}} + \frac{h^2}{\sqrt{h^2 + k^2}} \leq 4\sqrt{h^2 + k^2}.$$

$$\text{Also } \lim_{(h,k) \rightarrow \mathbf{0}} \sqrt{h^2 + k^2} = 0.$$

$$\text{Since } 0 \leq \lim_{(h,k) \rightarrow \mathbf{0}} \frac{|f(2+h, 1+k) - f(2, 1) - [\nabla f]_{(2,1)} \cdot \langle h, k \rangle|}{\sqrt{h^2 + k^2}} \leq 4 \lim_{(h,k) \rightarrow \mathbf{0}} \sqrt{h^2 + k^2} = 0,$$

$$\lim_{(h,k) \rightarrow \mathbf{0}} \frac{|f(2+h, 1+k) - f(2, 1) - [\nabla f]_{(2,1)} \cdot \langle h, k \rangle|}{\sqrt{h^2 + k^2}} = 0.$$

Hence f is differentiable at $(2, 1)$.

Aliter: Note that for all (x, y) such that $x + y \neq 0$, $f_x(x, y) = \frac{y}{(x+y)^2}$ and

$f_y(x, y) = -\frac{x}{(x+y)^2}$ are continuous functions, hence f_x, f_y are continuous throughout some small neighborhood of $(2, 1)$, hence f is differentiable at $(2, 1)$.

If $x = y \neq 0$, then $f(x, x) = \frac{x}{x+x} = \frac{1}{2}$, hence $\lim_{x \rightarrow 0} f(x, x) = \frac{1}{2}$, but $f(0, 0) = 0$, hence f is not continuous at $(0, 0)$, which implies f is not differentiable at $(0, 0)$.

(b) Note that for all (x, y) such that $x + e^y > 0$, $f_x(x, y) = \frac{1}{2\sqrt{x + e^y}}$ and

$f_y(x, y) = \frac{e^y}{2\sqrt{x + e^y}}$ are continuous, hence f_x, f_y are continuous throughout some small neighborhood of $(3, 0)$, hence f is differentiable at $(3, 0)$.

15. Show that the following function is differentiable throughout $\mathbf{R}^2$ and find the maximum rate of change of $f(x, y) = 6 - 3x^2 - y^2$ at the point $(1, 2)$ and the direction in which it occurs.

Solution: If f differentiable throughout $\mathbf{R}^2$, then the directional derivatives exist along any direction $\mathbf{u}$ at $(1, 2)$.

Then the directional derivative of f along any unit vector $\mathbf{u} = \langle u_1, u_2 \rangle$ at $(1, 2)$, is given by

$$D_{\mathbf{u}}f(1, 2) = [\nabla f]_{(1,2)} \cdot \langle u_1, u_2 \rangle = |[\nabla f]_{(1,2)}| \cos \theta, \text{ where } \theta \text{ is the angle between } [\nabla f]_{(1,2)} \text{ and } \mathbf{u}.$$

Hence the directional derivative is maximum when $\cos \theta = 1$, or $\theta = 0$ or $\mathbf{u}$ is parallel to and in the same direction as $[\nabla f]_{(1,2)}$.

$$\text{Hence the required } \mathbf{u} = |[\nabla f]_{(1,2)}| = \left\langle \frac{-6}{\sqrt{52}}, \frac{-4}{\sqrt{52}} \right\rangle.$$

To show that f is differentiable:

Check that for $(x, y) \in \mathbf{R}^2$, $f_x(x, y) = -6x$ and $f_y(x, y) = -2y$ are continuous func-

tions, hence f is differentiable throughout $\mathbf{R}^2$. (*)

((*) For any $(x_0, y_0) \in \mathbf{R}^2$

$$|f_x(x, y) - f_x(x_0, y_0)| = 6|x - x_0| \leq 6\sqrt{(x - x_0)^2 + (y - y_0)^2}.$$

Hence given $\epsilon > 0$ take $\delta = \frac{\epsilon}{6}$,

then $|f_x(x, y) - f_x(x_0, y_0)| < \epsilon$ if $\sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$

which implies f_x is continuous at (x_0, y_0) .

Similarly one can show that f_y is continuous at (x_0, y_0) .)

Aliter: f is differentiable at (x_0, y_0) if

$$\lim_{(h,k) \rightarrow \mathbf{0}} \frac{|f(x_0 + h, y_0 + k) - f(x_0, y_0) - [\nabla f]_{(x_0, y_0)} \cdot \langle h, k \rangle|}{\sqrt{h^2 + k^2}} = 0.$$

$$\begin{aligned} \frac{|f(x_0 + h, y_0 + k) - f(x_0, y_0) - [\nabla f]_{(x_0, y_0)} \cdot \langle h, k \rangle|}{\sqrt{h^2 + k^2}} &= \frac{|-3h^2 - k^2|}{\sqrt{h^2 + k^2}} = 3 \frac{h^2}{\sqrt{h^2 + k^2}} + \frac{k^2}{\sqrt{h^2 + k^2}} \\ &\leq 3\sqrt{h^2 + k^2} + \sqrt{h^2 + k^2} \text{ (since } h^2, k^2 \leq h^2 + k^2 \text{).} \\ &= 4\sqrt{h^2 + k^2}. \end{aligned}$$

$$\text{Also } \lim_{(h,k) \rightarrow \mathbf{0}} \sqrt{h^2 + k^2} = 0.$$

$$\text{Since } 0 \leq \lim_{(h,k) \rightarrow \mathbf{0}} \frac{|f(x_0 + h, y_0 + k) - f(x_0, y_0) - [\nabla f]_{(x_0, y_0)} \cdot \langle h, k \rangle|}{\sqrt{h^2 + k^2}} \leq 4 \lim_{(h,k) \rightarrow \mathbf{0}} \sqrt{h^2 + k^2} = 0,$$

$$\lim_{(h,k) \rightarrow \mathbf{0}} \frac{|f(x_0 + h, y_0 + k) - f(x_0, y_0) - [\nabla f]_{(x_0, y_0)} \cdot \langle h, k \rangle|}{\sqrt{h^2 + k^2}} = 0.$$

16. If R is the total resistance of three resistors, connected in parallel, with resistances R_1, R_2, R_3 , then

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

The resistances are measured in ohms as $R_1 = 100\Omega$, $R_2 = 100\Omega$ and $R_3 = 200\Omega$ at a particular instant. At any instant R_1 and R_2 are increasing at $1\Omega/s$ whereas R_3 is decreasing at $2\Omega/s$. Is R increasing or decreasing at that instant? At what rate?

Solution: At any time point t

$$\begin{aligned} \frac{-1}{R(t)^2} \frac{\partial R(t)}{\partial t} &= \frac{-1}{R_1(t)^2} \frac{\partial R_1(t)}{\partial t} + \frac{-1}{R_2(t)^2} \frac{\partial R_2(t)}{\partial t} + \frac{-1}{R_3(t)^2} \frac{\partial R_3(t)}{\partial t}. \\ \Rightarrow \frac{\partial R(t)}{\partial t} \left(\frac{1}{100} + \frac{1}{100} + \frac{1}{200} \right)^2 &= \left(\frac{1}{100^2}(1) + \frac{1}{100^2}(1) + \frac{1}{200^2}(-2) \right) \\ \Rightarrow \frac{\partial R(t)}{\partial t} &= \frac{4}{25} \times \frac{3}{2} = .24\Omega/s. \end{aligned}$$

17. Assume that $w = f(x, y)$, $x = r \cos \theta$ and $y = r \sin \theta$. Assuming the existence of all the required first and second order partial derivatives of w with respect to x, y, r and θ , show that

$$\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} = \frac{\partial^2 w}{\partial r^2} + \frac{1}{r} \frac{\partial w}{\partial r} + \frac{1}{r^2} \frac{\partial^2 w}{\partial \theta^2}.$$

Solution: Since $w = f(x, y) = f(r \cos \theta, r \sin \theta) = g(r, \theta)$.

By the chain rule we get

$$\frac{\partial g}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r}$$

But we will write $\frac{\partial g}{\partial r}$ as $\frac{\partial w}{\partial r}$.

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \cos \theta + \frac{\partial w}{\partial y} \sin \theta. \quad (1)$$

$$\text{Hence } \frac{\partial^2 g}{\partial r^2} = \cos \theta \left(\frac{\partial}{\partial r} \left(\frac{\partial w}{\partial x} \right) \right) + \sin \theta \left(\frac{\partial}{\partial r} \left(\frac{\partial w}{\partial y} \right) \right)$$

But we will write $\frac{\partial^2 g}{\partial r^2}$ as $\frac{\partial^2 w}{\partial r^2}$.

By again applying the chain rule we get

$$\begin{aligned} \frac{\partial^2 w}{\partial r^2} &= \cos \theta \left(\left(\frac{\partial}{\partial x} \left(\frac{\partial w}{\partial x} \right) \right) \frac{\partial x}{\partial r} + \left(\frac{\partial}{\partial y} \left(\frac{\partial w}{\partial x} \right) \right) \frac{\partial y}{\partial r} \right) + \sin \theta \left(\left(\frac{\partial}{\partial x} \left(\frac{\partial w}{\partial y} \right) \right) \frac{\partial x}{\partial r} + \left(\frac{\partial}{\partial y} \left(\frac{\partial w}{\partial y} \right) \right) \frac{\partial y}{\partial r} \right) \\ &= \left(\cos^2 \theta \frac{\partial^2 w}{\partial x^2} + \cos \theta \sin \theta \frac{\partial^2 w}{\partial y \partial x} \right) + \left(\sin \theta \cos \theta \frac{\partial^2 w}{\partial x \partial y} + \sin^2 \theta \frac{\partial^2 w}{\partial y^2} \right). \end{aligned} \quad (2)$$

By applying the chain rule we get

$$\begin{aligned} \frac{\partial g}{\partial \theta} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial \theta} \\ &= \frac{\partial w}{\partial x} (-r \sin \theta) + \frac{\partial w}{\partial y} (r \cos \theta). \end{aligned}$$

Again by applying the chain rule we get

$$\begin{aligned} \frac{\partial^2 g}{\partial \theta^2} &= \frac{\partial w}{\partial x} (-r \cos \theta) + (-r \sin \theta) \left(\frac{\partial^2 w}{\partial x^2} \frac{\partial x}{\partial \theta} + \frac{\partial^2 w}{\partial y \partial x} \frac{\partial y}{\partial \theta} \right) + \frac{\partial w}{\partial y} (-r \sin \theta) + \\ &+ (r \cos \theta) \left(\frac{\partial^2 w}{\partial y^2} \frac{\partial y}{\partial \theta} + \frac{\partial^2 w}{\partial x \partial y} \frac{\partial x}{\partial \theta} \right). \end{aligned}$$

But we will write $\frac{\partial^2 g}{\partial \theta^2}$ as $\frac{\partial^2 w}{\partial \theta^2}$.

$$\begin{aligned} \frac{\partial^2 w}{\partial \theta^2} &= (-r \cos \theta) \frac{\partial w}{\partial x} + (-r \sin \theta)^2 \left(\frac{\partial^2 w}{\partial x^2} \right) + (-r \sin \theta)(r \cos \theta) \left(\frac{\partial^2 w}{\partial y \partial x} \right) + (-r \sin \theta) \frac{\partial w}{\partial y} + \\ &+ (r \cos \theta)^2 \left(\frac{\partial^2 w}{\partial y^2} \right) + (r \cos \theta)(-r \sin \theta) \left(\frac{\partial^2 w}{\partial x \partial y} \right). \end{aligned} \quad (3)$$

From (1), (2) and (3) it follows that

$$\frac{\partial^2 w}{\partial r^2} + \frac{1}{r} \frac{\partial w}{\partial r} + \frac{1}{r^2} \frac{\partial^2 w}{\partial \theta^2} = \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}.$$

18. Suppose $w = f(u)$ where $u = \frac{x^2 - y^2}{x^2 + y^2}$. Assuming the existence of all the required first order partial derivatives of w and u show that $xw_x + yw_y = 0$.

Solution: By applying chain rule we get

$$w_x = \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} \text{ and } w_y = \frac{\partial f}{\partial u} \frac{\partial u}{\partial y}.$$

$$\frac{\partial u}{\partial x}|_{(x,y) \neq (0,0)} = \frac{4xy^2}{(x^2 + y^2)^2} \text{ and } \frac{\partial u}{\partial y}|_{(x,y) \neq (0,0)} = \frac{-4yx^2}{(x^2 + y^2)^2}.$$

$$\text{Hence } xw_x + yw_y = \frac{4x^2y^2}{(x^2 + y^2)^2} \frac{\partial f}{\partial u} + \frac{-4x^2y^2}{(x^2 + y^2)^2} \frac{\partial f}{\partial u} = 0.$$

19. **Implicit differentiation:** If $\phi(x, y, z) = 0$ defines z as an implicit function of x and y in a region R of the xy -plane, assuming the existence of all the required partial derivatives prove that $\frac{\partial z}{\partial x} = -\frac{\phi_x}{\phi_z}$ and $\frac{\partial z}{\partial y} = -\frac{\phi_y}{\phi_z}$, where $\phi_z \neq 0$. Hence find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ when $x^{\frac{2}{3}} + y^{\frac{2}{3}} + z^{\frac{2}{3}} = 1$.

Solution: Since $\phi(x, y, z) = 0$, gives z implicitly as a function of x and y we write $z = f(x, y)$. Hence $\phi(x, y, f(x, y)) = g(x, y) = 0$.

If we take $u_1(x, y) = x$, $u_2(x, y) = y$, and $u_3(x, y) = f(x, y) = z$, then we get,

$$\phi(u_1(x, y), u_2(x, y), u_3(x, y)) = g(x, y) = 0.$$

By applying chain rule we get

$$\frac{\partial \phi}{\partial u_1} \frac{\partial u_1}{\partial x} + \frac{\partial \phi}{\partial u_2} \frac{\partial u_2}{\partial x} + \frac{\partial \phi}{\partial u_3} \frac{\partial u_3}{\partial x} = \frac{\partial g}{\partial x} = 0, \text{ or}$$

$$\begin{aligned} \frac{\partial \phi}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial \phi}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial \phi}{\partial z} \frac{\partial z}{\partial x} &= 0. \\ \Rightarrow \frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial z} \frac{\partial z}{\partial x} &= 0 \text{ (since } \frac{\partial y}{\partial x} = 0). \end{aligned}$$

$$\text{Hence } \frac{\partial z}{\partial x} = -\frac{\phi_x}{\phi_z} \text{ when } \phi_z \neq 0,$$

$$\text{and similarly } \frac{\partial z}{\partial y} = -\frac{\phi_y}{\phi_z}, \text{ when } \phi_z \neq 0.$$

$$\text{If } \phi(x, y, z) = x^{\frac{2}{3}} + y^{\frac{2}{3}} + z^{\frac{2}{3}} - 1$$

$$\text{then } \phi_x = \frac{2}{3}x^{-\frac{1}{3}}, \phi_y = \frac{2}{3}y^{-\frac{1}{3}}, \phi_z = \frac{2}{3}z^{-\frac{1}{3}}.$$

$$\text{Hence } \frac{\partial z}{\partial x} = -\frac{\phi_x}{\phi_z} = -\left(\frac{z}{x}\right)^{\frac{1}{3}} \text{ and } \frac{\partial z}{\partial y} = -\frac{\phi_y}{\phi_z} = -\left(\frac{z}{y}\right)^{\frac{1}{3}}.$$

20. Suppose that $w = \frac{1}{r}f\left(t - \frac{r}{a}\right)$ and that $r = \sqrt{x^2 + y^2 + z^2}$. Assuming the existence of all the required second order partial derivatives, show that

$$\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} = \frac{1}{a^2} \frac{\partial^2 w}{\partial t^2}.$$

Solution: $w = \frac{1}{r}f\left(t - \frac{r}{a}\right) = g(x, y, z)$, say.

By applying the chain rule we get

$$\begin{aligned}
\frac{\partial g}{\partial x} &= \frac{\partial w}{\partial r} \frac{\partial r}{\partial x} \\
\Rightarrow \frac{\partial^2 g}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial w}{\partial r} \frac{\partial r}{\partial x} \right) \\
&= \left(\frac{\partial r}{\partial x} \right) \left(\frac{\partial}{\partial x} \left(\frac{\partial w}{\partial r} \right) \right) + \left(\frac{\partial w}{\partial r} \right) \left(\frac{\partial}{\partial x} \left(\frac{\partial r}{\partial x} \right) \right) \\
&= \frac{\partial r}{\partial x} \left(\left(\frac{\partial}{\partial r} \left(\frac{\partial w}{\partial r} \right) \right) \frac{\partial r}{\partial x} \right) + \frac{\partial w}{\partial r} \left(\frac{\partial^2 r}{\partial x^2} \right) \\
&= \left(\frac{\partial^2 w}{\partial r^2} \right) \left(\frac{\partial r}{\partial x} \right)^2 + \left(\frac{\partial w}{\partial r} \right) \left(\frac{\partial^2 r}{\partial x^2} \right).
\end{aligned}$$

Similarly $\frac{\partial^2 g}{\partial y^2} = \left(\frac{\partial^2 w}{\partial r^2} \right) \left(\frac{\partial r}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial r} \right) \left(\frac{\partial^2 r}{\partial y^2} \right)$,

and $\frac{\partial^2 g}{\partial z^2} = \left(\frac{\partial^2 w}{\partial r^2} \right) \left(\frac{\partial r}{\partial z} \right)^2 + \left(\frac{\partial w}{\partial r} \right) \left(\frac{\partial^2 r}{\partial z^2} \right)$.

$$\Rightarrow \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} = \left(\frac{\partial^2 w}{\partial r^2} \right) \left(\left(\frac{\partial r}{\partial x} \right)^2 + \left(\frac{\partial r}{\partial y} \right)^2 + \left(\frac{\partial r}{\partial z} \right)^2 \right) + \left(\frac{\partial w}{\partial r} \right) \left(\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} + \frac{\partial^2 r}{\partial z^2} \right).$$

Note that $\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2 + y^2 + z^2}}$, $\frac{\partial^2 r}{\partial x^2} = \frac{y^2 + z^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$.

Similarly $\frac{\partial r}{\partial y} = \frac{y}{\sqrt{x^2 + y^2 + z^2}}$, $\frac{\partial^2 r}{\partial y^2} = \frac{x^2 + z^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$

and $\frac{\partial r}{\partial z} = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$, $\frac{\partial^2 r}{\partial z^2} = \frac{x^2 + y^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$.

$$\Rightarrow \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} + \frac{\partial^2 g}{\partial z^2} = \left(\frac{\partial^2 w}{\partial r^2} \right) + \left(\frac{\partial w}{\partial r} \right) \frac{2}{r}. \quad (1)$$

Note that $\frac{\partial w}{\partial r} = -\frac{1}{r^2} f(u) + \frac{1}{r} \frac{\partial f(u)}{\partial u} \frac{\partial u}{\partial r}$

$$= -\frac{1}{r^2} f(u) + \frac{1}{r} f'(u) \left(-\frac{1}{a} \right), \text{ where } u = t - \frac{r}{a}, \text{ and}$$

$$\frac{\partial^2 w}{\partial r^2} = \left(\frac{2}{r^3} f(u) + \left(-\frac{1}{r^2} \right) f'(u) \left(-\frac{1}{a} \right) \right) + \left(\left(-\frac{1}{r^2} \right) f'(u) \left(-\frac{1}{a} \right) + \left(\frac{1}{r} \right) \left(-\frac{1}{a} \right) f''(u) \left(-\frac{1}{a} \right) \right),$$

where $u = t - \frac{r}{a}$.

Hence $\left(\frac{\partial^2 w}{\partial r^2} \right) + \left(\frac{\partial w}{\partial r} \right) \frac{2}{r} = \frac{1}{r} \left(-\frac{1}{a} \right) f''(u) \left(-\frac{1}{a} \right) = \frac{1}{a^2 r} f''(u)$. (2)

Also $\frac{\partial w}{\partial t} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial t} = \frac{1}{r} f'(u) \times 1$, where $u = t - \frac{r}{a}$ and

$$\frac{\partial^2 w}{\partial t^2} = \left(\frac{\partial}{\partial u} \left(\frac{1}{r} f'(u) \right) \right) \frac{\partial u}{\partial t} = \frac{1}{r} f''(u).$$

$$\Rightarrow \frac{1}{a^2 r} f''(u) = \frac{1}{a^2} \frac{\partial^2 w}{\partial t^2}. \quad (3)$$

(1), (2) and (3) $\Rightarrow \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} = \frac{1}{a^2} \frac{\partial^2 w}{\partial t^2}$.

(We write $\frac{\partial^2 g}{\partial x^2}$, $\frac{\partial^2 g}{\partial y^2}$ and $\frac{\partial^2 g}{\partial z^2}$ as $\frac{\partial^2 w}{\partial x^2}$, $\frac{\partial^2 w}{\partial y^2}$ and $\frac{\partial^2 w}{\partial z^2}$, respectively).

21. A function f is called **homogeneous of degree n** if it satisfies the equation $f(tx, ty) = t^n f(x, y)$ for all t , where n is a positive integer and f has continuous second order partial derivatives.

(a) Verify that $f(x, y) = x^3 - 2xy^2 + 5y^3$ is homogeneous of degree 3.

Solution: Can be easily worked out.

(b) Show that if f is homogeneous of degree n , then

- i. $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf(x, y),$
- ii. $x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = n(n-1)f(x, y)$
- iii. $f_x(tx, ty) = t^{n-1}f_x(x, y).$

Solution:

(i) $f(tx, ty) = t^n f(x, y)$ for all $t \in \mathbf{R}$

$$\frac{\partial f(u, v)}{\partial t} = \frac{\partial f(u, v)}{\partial u} \frac{\partial u}{\partial t} + \frac{\partial f(u, v)}{\partial v} \frac{\partial v}{\partial t} = nt^{n-1}f(x, y) \text{ for all } t \in \mathbf{R},$$

where $u = tx$ and $v = ty$.

$$\Rightarrow \frac{\partial f(u, v)}{\partial t} = \frac{\partial f(u, v)}{\partial u} x + \frac{\partial f(u, v)}{\partial v} y = nt^{n-1}f(x, y). \quad (1)$$

Hence for $t = 1$ or $u = x$ and $v = y$ we get:

$$\Rightarrow \frac{\partial f(x, y)}{\partial x} x + \frac{\partial f(x, y)}{\partial y} y = nf(x, y).$$

(ii)

By taking the partial derivative with respect to t on both sides of equation (1) we get

$$\begin{aligned} \frac{\partial^2 f(u, v)}{\partial t^2} &= x \left(\left(\frac{\partial}{\partial u} \left(\frac{\partial f(u, v)}{\partial u} \right) \right) \frac{\partial u}{\partial t} + \left(\frac{\partial}{\partial v} \left(\frac{\partial f(u, v)}{\partial u} \right) \right) \frac{\partial v}{\partial t} \right) + \\ &y \left(\left(\frac{\partial}{\partial v} \left(\frac{\partial f(u, v)}{\partial v} \right) \right) \frac{\partial v}{\partial t} + \left(\frac{\partial}{\partial u} \left(\frac{\partial f(u, v)}{\partial v} \right) \right) \frac{\partial u}{\partial t} \right) \\ &= n(n-1)t^{n-2}f(x, y) \text{ for all } t \in \mathbf{R}, \text{ where } u = tx \text{ and } v = ty. \end{aligned} \quad (2)$$

$$(2) \Rightarrow x^2 \frac{\partial^2 f(u, v)}{\partial u^2} + 2xy \frac{\partial^2 f(u, v)}{\partial u \partial v} + y^2 \frac{\partial^2 f(u, v)}{\partial v^2} = n(n-1)t^{n-2}f(x, y)$$

$$(\text{since } \frac{\partial^2 f(u, v)}{\partial u \partial v} = \frac{\partial^2 f(u, v)}{\partial v \partial u}).$$

Hence for $t = 1$ or $u = x$ and $v = y$ we get:

$$x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = n(n-1)f(x, y).$$

$$\begin{aligned} \text{(iii) } f_x(tx, ty) &= \lim_{h \rightarrow 0} \frac{f(tx + th, ty) - f(tx, ty)}{th} \\ &= \lim_{h \rightarrow 0} \frac{t^n(f(x + h, y) - f(x, y))}{th} \\ &= t^{n-1} \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h} = t^{n-1}f_x(x, y). \end{aligned}$$